COMP3602 - Week 1 Extension Activity

Sales Data Investigation | Selection, Filtering, Querying & Visualization

Scenario
You are given a synthetic sales dataset containing 240 orders from five locations. Work through the missions sequentially and use Pandas/Matplotlib to answer increasingly open-ended questions.

How to work
Complete the CORE questions first. If you finish early, attempt the GO FURTHER challenge. In the final missions, decide what the visualization tells you rather than only producing a chart.

Mission 1 - LOAD and meet the DataFrame

```
import pandas as pd
sales = pd.read_csv("COMP3602_Week1_Sales_Data.csv")
```

Download COMP3602_Week1_Sales_Data.csv from Moodle and load it into Pandas.

1. Display the first five rows.

2. How many rows and columns are in the dataset?

3. Display the column names.

4. Use info() to inspect the dataset. Which columns appear numerical? Which appear categorical/text?

GO FURTHER
Find a Pandas command that gives a statistical summary of the numerical columns.

Mission 2 - Series vs DataFrame

5. Select only the Revenue column. What type of Pandas object is returned?

6. Select Region, Product and Revenue together. What type of Pandas object is returned?

7. Display the first five values of the Revenue Series.

GO FURTHER
Find the average, minimum and maximum values of the Revenue Series.

Mission 3 - Selecting data with loc and iloc

8. Using iloc, display the first row.

9. Using iloc, display the first five rows and the first four columns.

10. Using loc, display rows 0 to 4 but only Order_ID, Region, Product and Revenue.

11. Using loc, retrieve the Region value at index 10.

GO FURTHER
Use iloc to retrieve the sixth order. Then use loc to display that complete row using its index.

Mission 4 - Filter the DataFrame

12. Show all orders from Muscat.

13. Show all Electronics orders.

14. Show all orders with Revenue greater than 500 OMR.

15. How many orders have Revenue greater than 500 OMR?

GO FURTHER
Find all orders that are BOTH from Muscat AND belong to Electronics. How many are there?

Mission 5 - Multiple conditions and query()

16. Find Business customer orders that received a discount of at least 10%.

17. Using query(), find orders from Muscat with Revenue greater than 500 OMR.

18. Compare this with normal Boolean filtering. Do both approaches return the same records?

GO FURTHER
Write ONE query() expression that returns Electronics orders with at least 5 units sold and Revenue above 250 OMR.

Mission 6 - VISUALIZE the data

Now move beyond rows and numbers. Use visualization to look for patterns that are harder to notice in a table.

19. Investigate whether orders with more Units tend to generate higher Revenue.

Create a scatter plot with Units on the x-axis and Revenue on the y-axis. Add a title and axis labels.

```
import matplotlib.pyplot as plt
```

```
plt.scatter(sales["Units"], sales["Revenue"])
plt.xlabel("Units Sold")
plt.ylabel("Revenue (OMR)")
plt.title("Units Sold vs Revenue")
plt.show()
```

20. What does one dot represent?

21. Describe the overall relationship you observe. Is it perfect? Why might orders with the same number of units have different revenue?

GO FURTHER - Investigate on your own
Choose ONE additional relationship or comparison in the sales data and visualize it appropriately. Examples: compare total Revenue across Regions or Categories, investigate Unit_Price vs Revenue, or examine how Revenue changes over time. Do not simply create a graph - write 1-2 sentences explaining what it reveals.

Mission 7 - Mini investigation

Use any techniques learned today: Series/DataFrame selection, loc/iloc, filtering, query() and visualization.

22. Identify the order with the highest Revenue. Report its Order_ID, Region, Product, Units and Revenue.

23. Which region generated the largest total Revenue? Create a suitable visualization to communicate the comparison.

24. Write one short evidence-based conclusion about this sales dataset.

GO FURTHER - Open Challenge
Which salesperson generated the highest total Revenue? Create a suitable visualization comparing salespeople and explain your finding. You may need to discover a Pandas operation that has not yet been demonstrated.

Focus: the chart is not the final answer. The important part is what you can reasonably interpret from it.